

Filtering and Sorting Assignment

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Assignment Name: Filtering and Sorting Assignment

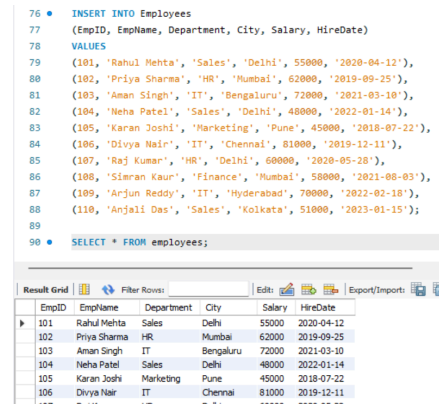
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Creating given dummy table in MySQL Workbench using given data :

```
CREATE TABLE Employees (  
    EmpID INT PRIMARY KEY,  
    EmpName VARCHAR(50),  
    Department VARCHAR(50),  
    City VARCHAR(50),  
    Salary INT,  
    HireDate DATE  
);
```

```
INSERT INTO Employees  
(EmpID, EmpName, Department, City, Salary, HireDate)  
VALUES  
(101, 'Rahul Mehta', 'Sales', 'Delhi', 55000, '2020-04-12'),  
(102, 'Priya Sharma', 'HR', 'Mumbai', 62000, '2019-09-25'),  
(103, 'Aman Singh', 'IT', 'Bengaluru', 72000, '2021-03-10'),  
(104, 'Neha Patel', 'Sales', 'Delhi', 48000, '2022-01-14'),  
(105, 'Karan Joshi', 'Marketing', 'Pune', 45000, '2018-07-22'),  
(106, 'Divya Nair', 'IT', 'Chennai', 81000, '2019-12-11'),  
(107, 'Raj Kumar', 'HR', 'Delhi', 60000, '2020-05-28'),  
(108, 'Simran Kaur', 'Finance', 'Mumbai', 58000, '2021-08-03'),  
(109, 'Arjun Reddy', 'IT', 'Hyderabad', 70000, '2022-02-18'),  
(110, 'Anjali Das', 'Sales', 'Kolkata', 51000, '2023-01-15');
```



```
76 • INSERT INTO Employees  
77 (EmpID, EmpName, Department, City, Salary, HireDate)  
78 VALUES  
79 (101, 'Rahul Mehta', 'Sales', 'Delhi', 55000, '2020-04-12'),  
80 (102, 'Priya Sharma', 'HR', 'Mumbai', 62000, '2019-09-25'),  
81 (103, 'Aman Singh', 'IT', 'Bengaluru', 72000, '2021-03-10'),  
82 (104, 'Neha Patel', 'Sales', 'Delhi', 48000, '2022-01-14'),  
83 (105, 'Karan Joshi', 'Marketing', 'Pune', 45000, '2018-07-22'),  
84 (106, 'Divya Nair', 'IT', 'Chennai', 81000, '2019-12-11'),  
85 (107, 'Raj Kumar', 'HR', 'Delhi', 60000, '2020-05-28'),  
86 (108, 'Simran Kaur', 'Finance', 'Mumbai', 58000, '2021-08-03'),  
87 (109, 'Arjun Reddy', 'IT', 'Hyderabad', 70000, '2022-02-18'),  
88 (110, 'Anjali Das', 'Sales', 'Kolkata', 51000, '2023-01-15');  
89  
90 • SELECT * FROM employees;
```

EmpID	EmpName	Department	City	Salary	HireDate
101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
102	Priya Sharma	HR	Mumbai	62000	2019-09-25
103	Aman Singh	IT	Bengaluru	72000	2021-03-10
104	Neha Patel	Sales	Delhi	48000	2022-01-14
105	Karan Joshi	Marketing	Pune	45000	2018-07-22
106	Divya Nair	IT	Chennai	81000	2019-12-11
107	Raj Kumar	HR	Delhi	60000	2020-05-28

Q1. Show employees working in either the 'IT' or 'HR' departments.

Answer:

```
SELECT *  
FROM Employees  
WHERE Department IN ('IT', 'HR');
```

```
92 • SELECT *
93 FROM Employees
94 WHERE Department IN ('IT', 'HR');
95
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmpID	EmpName	Department	City	Salary	HireDate
▶	102	Priya Sharma	HR	Mumbai	62000	2019-09-25
	103	Aman Singh	IT	Bengaluru	72000	2021-03-10
	106	Divya Nair	IT	Chennai	81000	2019-12-11
	107	Raj Kumar	HR	Delhi	60000	2020-05-28
	109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
✱	NULL	NULL	NULL	NULL	NULL	NULL

Q2. Retrieve employees whose department is 'Sales', 'IT', or 'Finance'.

Answer:

```
SELECT *  
FROM Employees  
WHERE Department IN ('Sales', 'IT', 'Finance');
```

```
96 • SELECT *
97 FROM Employees
98 WHERE Department IN ('Sales', 'IT', 'Finance');
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmpID	EmpName	Department	City	Salary	HireDate
▶	101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
	103	Aman Singh	IT	Bengaluru	72000	2021-03-10
	104	Neha Patel	Sales	Delhi	48000	2022-01-14
	106	Divya Nair	IT	Chennai	81000	2019-12-11
	108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
	109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
	110	Anjali Das	Sales	Kolkata	51000	2023-01-15
✱	NULL	NULL	NULL	NULL	NULL	NULL

Q3. Display employees whose salary is between ₹50,000 and ₹70,000.

Answer:

```
SELECT *  
FROM Employees  
WHERE Salary BETWEEN 50000 AND 70000;
```

```
100 • SELECT *
101 FROM Employees
102 WHERE Salary BETWEEN 50000 AND 70000;
103
```

Result Grid

Filter Rows:

Edit:

Export/Import:

Wrap Cell Content:

	EmpID	EmpName	Department	City	Salary	HireDate
▶	101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
	102	Priya Sharma	HR	Mumbai	62000	2019-09-25
	107	Raj Kumar	HR	Delhi	60000	2020-05-28
	108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
	109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
	110	Anjali Das	Sales	Kolkata	51000	2023-01-15
*	NULL	NULL	NULL	NULL	NULL	NULL

Q4. List employees whose names start with the letter 'A'.

Answer:

```
SELECT *  
FROM Employees  
WHERE EmpName LIKE 'A%';
```

```
104 • SELECT *
105 FROM Employees
106 WHERE EmpName LIKE 'A%';
107
```

Result Grid

  Filter Rows:

Edit:   

Export/Import:  

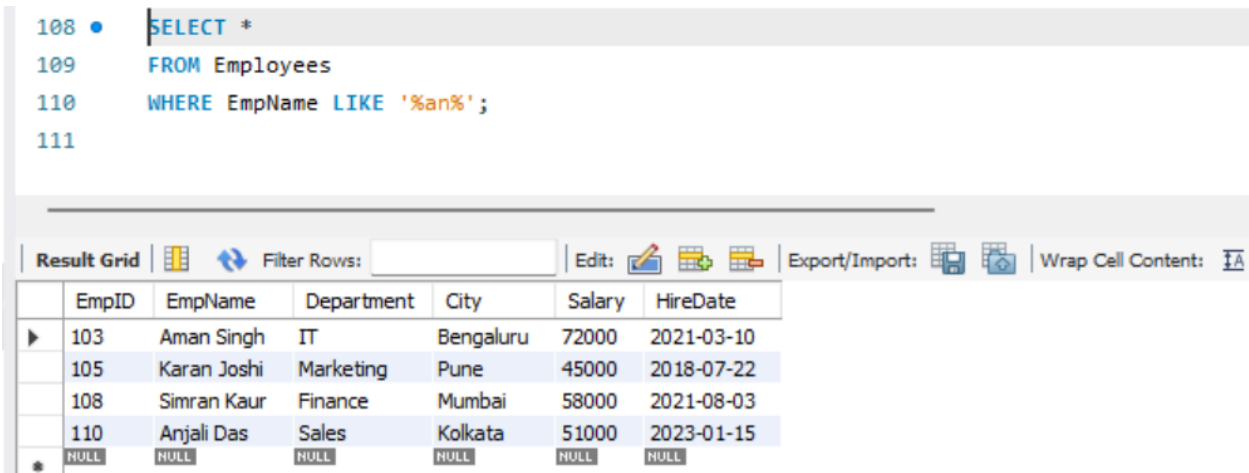
Wrap Cell Content: 

	EmpID	EmpName	Department	City	Salary	HireDate
▶	103	Aman Singh	IT	Bengaluru	72000	2021-03-10
	109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
	110	Anjali Das	Sales	Kolkata	51000	2023-01-15
*	NULL	NULL	NULL	NULL	NULL	NULL

Q5. Find employees whose names contain the substring 'an'.

Answer:

```
SELECT *  
FROM Employees  
WHERE EmpName LIKE '%an%';
```



The screenshot shows a SQL query editor with the following query:

```
108 • SELECT *  
109 FROM Employees  
110 WHERE EmpName LIKE '%an%';  
111
```

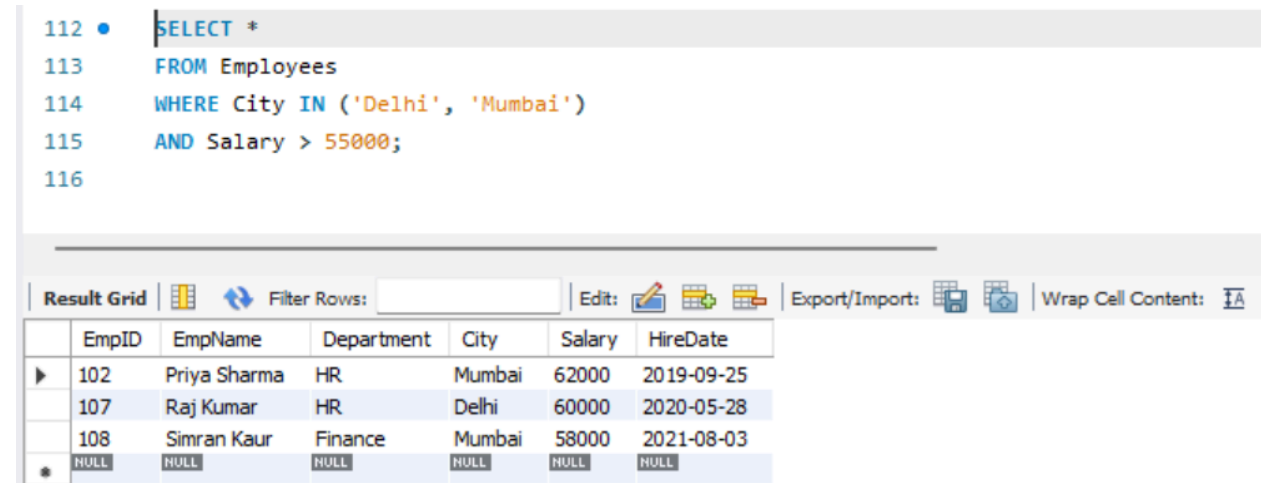
Below the query editor is a toolbar with options: Result Grid, Filter Rows, Edit, Export/Import, and Wrap Cell Content. The results are displayed in a table:

	EmpID	EmpName	Department	City	Salary	HireDate
▶	103	Aman Singh	IT	Bengaluru	72000	2021-03-10
	105	Karan Joshi	Marketing	Pune	45000	2018-07-22
	108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
	110	Anjali Das	Sales	Kolkata	51000	2023-01-15
*	NULL	NULL	NULL	NULL	NULL	NULL

Q6. Show employees who are from 'Delhi' or 'Mumbai' and earn more than ₹55,000.

Answer:

```
SELECT *  
FROM Employees  
WHERE City IN ('Delhi', 'Mumbai')  
AND Salary > 55000;
```



The screenshot shows a SQL query editor with the following query:

```
112 • SELECT *  
113 FROM Employees  
114 WHERE City IN ('Delhi', 'Mumbai')  
115 AND Salary > 55000;  
116
```

Below the query editor is a toolbar with options: Result Grid, Filter Rows, Edit, Export/Import, and Wrap Cell Content. The results are displayed in a table:

	EmpID	EmpName	Department	City	Salary	HireDate
▶	102	Priya Sharma	HR	Mumbai	62000	2019-09-25
	107	Raj Kumar	HR	Delhi	60000	2020-05-28
	108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
*	NULL	NULL	NULL	NULL	NULL	NULL

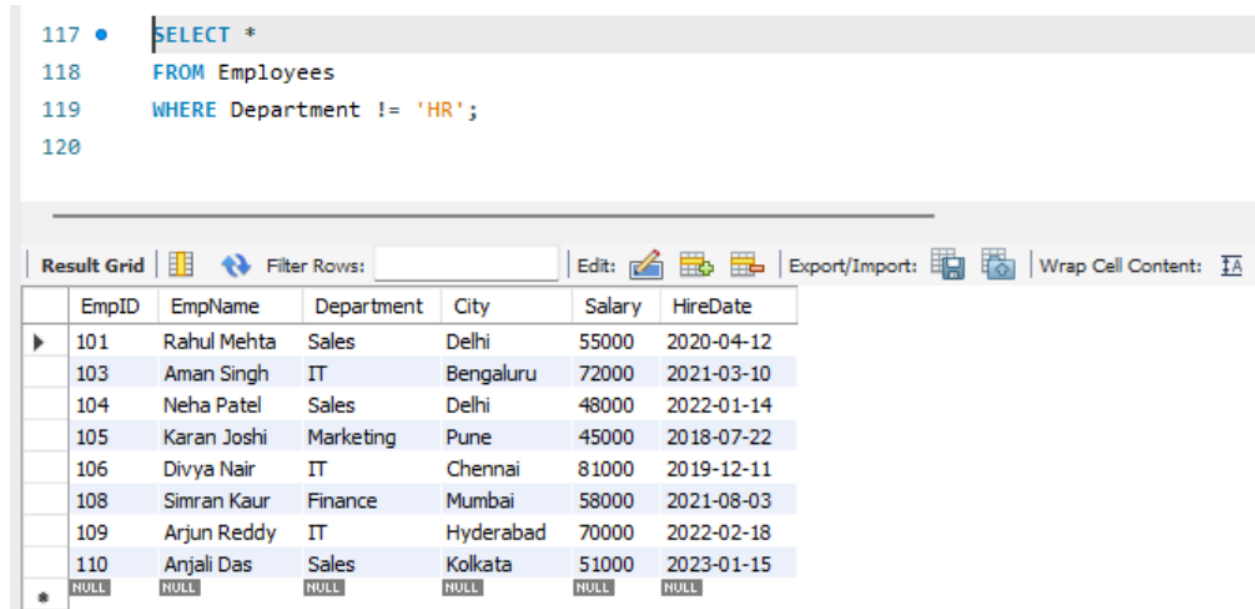
Q7. Display all employees except those from the 'HR' department.

Answer:

```
SELECT *
```

```
FROM Employees
```

```
WHERE Department != 'HR';
```



```
117 • SELECT *
118 FROM Employees
119 WHERE Department != 'HR';
120
```

	EmpID	EmpName	Department	City	Salary	HireDate
▶	101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
	103	Aman Singh	IT	Bengaluru	72000	2021-03-10
	104	Neha Patel	Sales	Delhi	48000	2022-01-14
	105	Karan Joshi	Marketing	Pune	45000	2018-07-22
	106	Divya Nair	IT	Chennai	81000	2019-12-11
	108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
	109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
	110	Anjali Das	Sales	Kolkata	51000	2023-01-15
*	NULL	NULL	NULL	NULL	NULL	NULL

Q8. Get all employees hired between 2019 and 2022, ordered by HireDate (Oldest First).

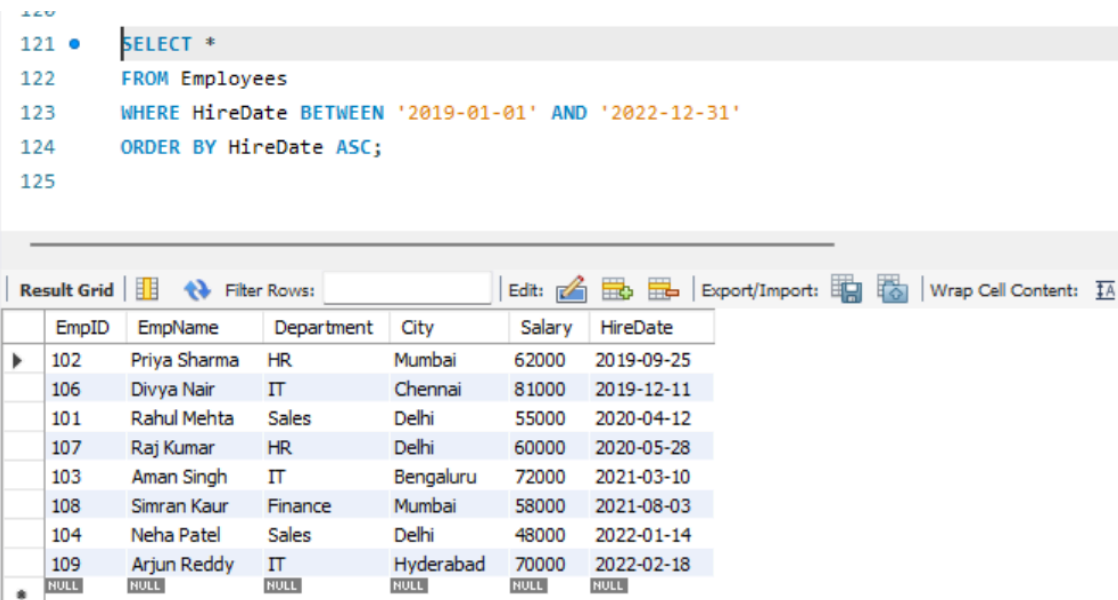
Answer:

```
SELECT *
```

```
FROM Employees
```

```
WHERE HireDate BETWEEN '2019-01-01' AND '2022-12-31'
```

```
ORDER BY HireDate ASC;
```



```
121 • SELECT *
122 FROM Employees
123 WHERE HireDate BETWEEN '2019-01-01' AND '2022-12-31'
124 ORDER BY HireDate ASC;
125
```

	EmpID	EmpName	Department	City	Salary	HireDate
▶	102	Priya Sharma	HR	Mumbai	62000	2019-09-25
	106	Divya Nair	IT	Chennai	81000	2019-12-11
	101	Rahul Mehta	Sales	Delhi	55000	2020-04-12
	107	Raj Kumar	HR	Delhi	60000	2020-05-28
	103	Aman Singh	IT	Bengaluru	72000	2021-03-10
	108	Simran Kaur	Finance	Mumbai	58000	2021-08-03
	104	Neha Patel	Sales	Delhi	48000	2022-01-14
	109	Arjun Reddy	IT	Hyderabad	70000	2022-02-18
*	NULL	NULL	NULL	NULL	NULL	NULL